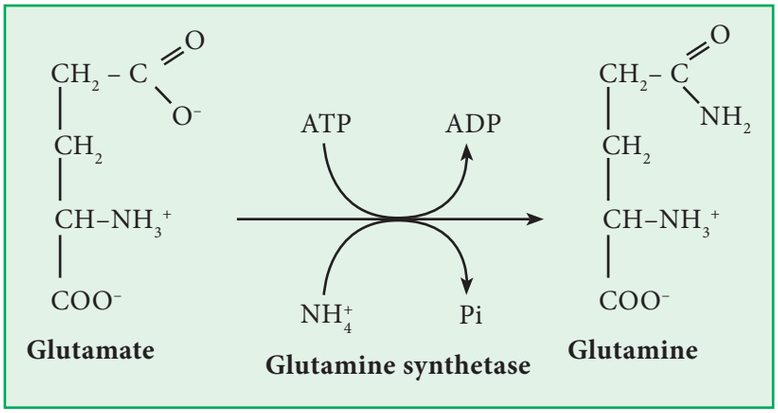
6) Ligases:

These enzymes catalyze the synthetic reactions. They link two substrates together with the utilization of ATP or GTP.

Example : Glutamine synthetase.

Glutamine synthetase (EC 6.3.1.2):

This is a ligase which catalyzes the synthesis of glutamine from glutamate and NH_3.



4.3 Coenzyme

1. Some of the enzymes are simple proteins. Examples: amylase, trypsin. Many enzymes require one or more non-protein components called cofactors. If the cofactor is an organic molecule, it is known as coenzyme. The cofactor may also be a metal ion.

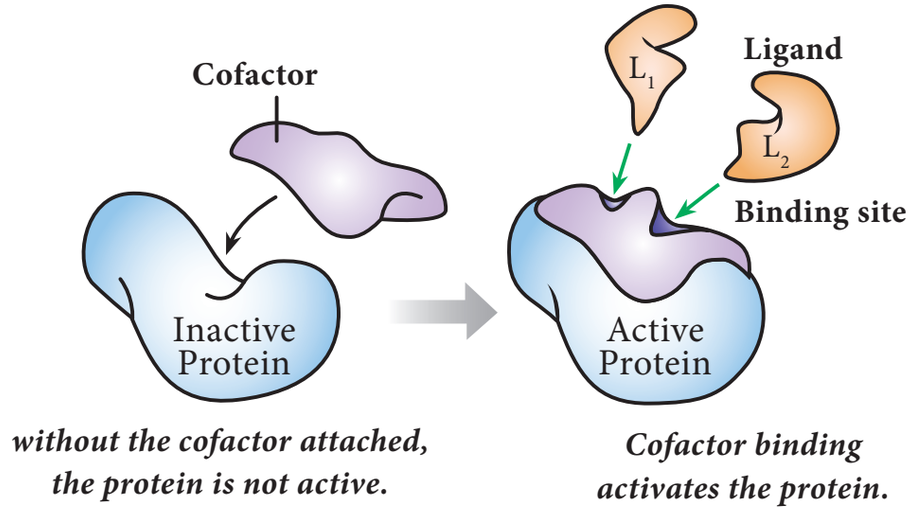


Figure 4.1 Cofactor binding

2. Coenzymes may be defined as non-protein, low molecular weight organic compounds required for the activity of enzymes. Example: Thiamine pyrophosphate (TPP).

Unit 4.indd 106 21-05-2018 15:10:06